

SOC Internship - Day 7

Topic: Alerts and Incident Response

1. What is an Alert?

An alert is a notification or warning generated by a security tool when suspicious, malicious, or unusual activity is detected in a system or network.

2. Types of Alerts

- Failed Login Attempts Alert
- Malware Detection Alert
- Unauthorized Access Alert
- Network Traffic Alert
- Phishing Email Alert

3. Alert Workflow

1. **Log Collection:** Security tools like SIEM, Firewall, IDS/IPS, EDR, and Antivirus collect logs and events from systems and networks.
2. **Detection and Correlation:** The security platform analyzes logs and correlates events to identify suspicious activity.
3. **Alert Generation:** If unusual behavior is detected, the system creates an alert.
4. **Alert Triage:** SOC analysts review the alert and determine its priority and severity.
5. **Investigation:** Analysts examine logs, affected systems, IP addresses, users, and related events.
6. **Response and Mitigation:** If the threat is confirmed, actions are taken such as blocking IPs, isolating devices, or removing malware.
7. **Closure and Documentation:** The incident is documented, lessons are recorded, and the alert is closed.

4. What is Incident Response?

Incident response is the process of identifying, managing, and fixing a cybersecurity incident such as a malware attack, phishing attack, data breach, or unauthorized access.

5. Incident Response Steps

8. **Preparation:** Get tools, plans, and teams ready before an attack happens.
9. **Detection and Analysis:** Find the incident and understand what happened.
10. **Containment:** Stop the threat from spreading. Example: isolate an infected computer.
11. **Eradication:** Remove the cause of the attack. Example: delete malware or block malicious IPs.
12. **Recovery:** Restore normal systems and services safely.

13. **Lessons Learned:** Review what happened and improve security for the future.

6. Example Scenario

A company's SOC uses a SIEM to monitor login activity. One day, the SIEM detects:

- 20 failed login attempts to an admin account in 2 minutes
- 1 successful login from the same IP address

This triggers an alert.

14. **Alert Generated:** The SIEM flags the activity as suspicious.

15. **SOC Analyst Reviews It:** The analyst checks the source IP address, username targeted, time of activity, and whether the IP is known or unusual.

16. **Investigation:** The analyst sees many failed attempts before the successful login. This may indicate a brute-force attack or credential compromise.

17. **Response:** The SOC team may block the IP address, force a password reset, disable the account temporarily, and check for other suspicious actions after login.

18. **Closure:** The incident is documented and the alert is closed after action is taken.